REFERENCE CARD · REV. 1

A5K — Technical Specifications

Clocks, memory, processors, bandwidth and the limits they impose

1 November 2026 · Companion to the design proposition · Figures are targets, not measurements

> **Design study.** Nothing described here has been built or measured. Clock rates are synthesis estimates for the named device families; bandwidth figures are arithmetic from bus width and data rate; processor core counts are what the logic can hold, not what the software needs.

1. Clock domains

Table S1. The domains a card would run in.

Domain	Rate	Purpose
Copper / chip bus	7.09 MHz (141 ns)	One instruction per tick; the machine's timing contract
Pixel clock, 1080p60	148.5 MHz	Scanout
Pixel clock, 4K60	594 MHz (4 lanes)	Scanout, above the fabric's comfort zone
DDR3L	800 MHz (1600 MT/s)	Chip RAM
PCIe reference	100 MHz	Link training and packet clock
Host CPU	external	Whatever the host provides

2. Memory

Table S2. Chip RAM configurations.

Configuration	Devices	Capacity	Bandwidth	Note
Base	2 × 4 Gb DDR3L, 32-bit	1 GB	6.4 GB/s	as costed in the BOM
Upgrade	2 × 8 Gb DDR3L, 32-bit	2 GB	6.4 GB/s	same package, drop-in
Maximum	4 × 8 Gb, two ranks	4 GB	6.4 GB/s	top of the 32-bit space
Architectural ceiling	32-bit byte addressing	4 GiB	—	~3.5 GB usable after registers and apertures
64-bit variant	4 × 16 Gb DDR4	8–32 GB	12.8–25 GB/s	wider bus and 40-bit pointers; new part

Why DDR3L, and not DDR5. The target families cannot drive DDR5 at all — it needs an UltraScale+ or Versal part, at roughly ten times the FPGA cost. DDR3L is still in production for industrial and networking equipment, is stocked by every distributor, and costs about $7 per 512 MB. A 32-bit DDR3L bus supplies 6.4 GB/s, which is more than the slot table can consume. DDR5 would solve a bandwidth problem this machine does not have, at a price it cannot pay.

Upgrade path. 1 GB to 2 GB is a device swap in the same footprint — 4 Gb parts for 8 Gb. 2 GB to 4 GB needs two ranks and rank support in the memory controller. Beyond 4 GiB the obstacle is not the memory but the pointer width: the 64-bit variant widens both, and moves to DDR4.

> **ECC** is the more valuable upgrade for instrumentation work than capacity: on a 32-bit bus it means 40 bits and five x8 devices instead of two x16. DDR3 tops out at 8 Gb per device, which is why 4 GB is the

> wall for the 32-bit design. Host memory is reached by DMA over PCIe and is not limited by the card.

3. Processors

There are two CPU positions in this design, and they are not interchangeable.

Table S3. Processor options, clock and cores.

Option	Kind	Max clock	Cores	Where it runs
VexRiscv on ECP5-85	soft	~50 MHz	1–4	On the card; housekeeping
VexRiscv on Artix-7 200T	soft	100–150 MHz	1–8	On the card; practical sweet spot
Microblaze on Artix-7	soft	~200 MHz	1–8	Vendor core, closed toolchain
Rocket / CVA6	soft	50–100 MHz	1–4	Needs a larger part
PolarFire SoC	hard	667 MHz	4 + 1	Hardened cluster beside the fabric
x86-64 host	hard	4–6 GHz	4–256	PC, outside the card
RISC-V host (P670 class)	hard	~2–2.5 GHz	1–16	Host board

> **The on-card core is a supervisor, not an engine.** It runs CIA timers, GPIO and interrupt aggregation. Whether it runs at 50 MHz or 667 MHz changes almost nothing, because it does not draw pixels — the Copper, blitter and planes do. Spending fabric on clock speed here is spending it in the wrong place. The heavy work belongs on the host, on the other side of PCIe.

4. Bandwidth budget at 4K60

3840 × 2160 = 8.29 megapixels. The 32-bit DDR3L bus supplies 6.4 GB/s, and the slot table must also fit the Copper, the blitter and audio into the same schedule.

Table S4. What 4K60 costs, per content type.

Content	Per frame	At 60 Hz	Verdict
8 planar bitplanes (1 bit/px)	8.3 MB	0.5 GB/s	Comfortable — 8% of budget
1 full-screen ARGB8888 plane	33 MB	2.0 GB/s	Comfortable
2 ARGB planes	66 MB	4.0 GB/s	Near the ceiling
4 ARGB planes	133 MB	8.0 GB/s	Over budget on 32-bit
8 ARGB planes	265 MB	15.9 GB/s	Needs the 64-bit variant

> This is the quiet argument for planar graphics: at 4K60, eight bitplanes cost a twelfth of what eight ARGB planes cost. The Amiga's bitmap format was a bandwidth strategy, not a limitation.

5. Parallelism

Table S5. What runs simultaneously, and what does not.

Unit	Count	Nature
Copper	1	One sequencer, one instruction per tick; parallel with everything else, serial within itself
Blitter	1	One operation at a time, by design
Planes	8 (16 optional)	Fetched in parallel each line
Audio channels	4–8	Independent DMA streams
Slot table	1	Fixed schedule; every cycle allocated before the frame
Host cores	unbounded	Outside the card, over PCIe

Parallelism here is spatial, not threaded. Roughly twenty independent units run at once because each does one thing forever at a fixed rate — not because anything is scheduled. Adding CPU cores does not add throughput; it adds contention on the same memory. The bound is bandwidth: at 6.4 GB/s, the display pipeline is the budget, and the slot table is the ledger.

6. Links and ceilings

Table S6. The buses that bound the design.

Link	Bandwidth	Binds
PCIe Gen2 ×4	2.0 GB/s	Host ↔ card, baseline
PCIe Gen3 ×4	3.9 GB/s	Host ↔ card, preferred
PCIe Gen4 ×4	7.9 GB/s	Host ↔ card, beyond current parts
DDR3L 32-bit	6.4 GB/s	All on-card fetching
Address space	4 GiB	Chip RAM ceiling

What would have to change for more. More planes at 4K means a wider memory bus before it means more logic. A faster on-card CPU means a hard core, not a bigger soft one. More host bandwidth means Gen3 or Gen4, which means an Artix-7 with GTP or a larger part. Each is a different board, not a different setting.

7. Quick reference

Table S7. The numbers worth memorising.

Item	Figure
Chip RAM, base	1 GB DDR3L, 32-bit, 1600 MT/s
Chip RAM, upgrade	2 GB — 8 Gb devices, same footprint
Chip RAM, maximum	4 GB, two ranks
Architectural ceiling	4 GiB, about 3.5 GB usable
Memory bandwidth	6.4 GB/s
Copper tick	7.09 MHz, 141 ns per instruction
Scanout	4K60 at a 594 MHz pixel clock, four lanes
Host link	PCIe Gen2 ×4 (2.0 GB/s), Gen3 ×4 (3.9 GB/s)
On-card CPU	VexRiscv, 50 MHz (ECP5) to 150 MHz (Artix-7)
Host CPU	anything with PCIe; 2–6 GHz, 4–256 cores
Planes	8, optionally 16
Blitters	1
Audio	4–8 DMA streams, 24-bit I²S
4K60 cost, planar	0.5 GB/s for eight bitplanes
4K60 cost, ARGB	2.0 GB/s per full-screen plane

> Sources: datasheet figures for the ECP5-85 and Artix-7 200T families, DDR3L at 1600 MT/s, and arithmetic from bus width and pixel format. No figure here has been measured on hardware, because there is no hardware.